

Comparative Electrical Modeling of Pinhole Defects with Copper Inclusion and Air-Filled void

Abstract—Through-silicon Vias (TSVs) serves as vertical interconnects in three dimensional integrated circuits (3D ICs), providing high integration density, reduced interconnect length and improved electrical performance. However, fabrication-induced defects, specifically weak or pinhole defects in the dielectric liner, can introduce additional parasitic effects that degrade TSV reliability and signal integrity. To address aforementioned issue, this paper presents the equivalent electrical modelling of a pinhole defected TSV. The proposed model consider two pinhole defect scenarios: a copper-leaked and an air void pinhole defect. The corresponding equivalent parasitic models are formulated by including defect. Subsequently, closed form analytical expression are derived to accurately characterized electrical behavior of the defected via. The equivalent models are implemented to evaluate performance metrics including noise, delay, crosstalk, leakage current and leakage power. Furthermore, the S parameter characteristics are investigated to assess the high frequency behavior and observed that losses is increased by 5.46%. This study provides a systematic framework for analyzing fabrication induced pinhole defect and evaluating their impact on the reliability and signal integrity of TSV based 3D ICs.

Index Terms—Pinhole Defect, Reliability, Signal Integrity, S-Parameters, Through-Silicon Via (TSV).

I. INTRODUCTION

THE continuous demand for higher performance, compact size, reduced functionality has driven significant advancements in integrated circuit technologies. Among the various emerging solutions, three-dimensional (3D) integration has gained considerable attention as an effective approach for overcoming the limitation of conventional planar integrated circuits (IC's). By enabling vertical stacking of multiple device layers within a single package, 3D IC have become an important platform for developing next-generation electronic system that require high speed, increased integration capability, and efficient power management[1]. Despite the continuous advancement in fabrication technology, fundamental limitations started to emerge as the dimensions of the TSV continued to shrink. Although TSV fabrication technology offers significant benefits for 3D integration, the fabrication process remains susceptible to manufacturing defects and process-induced variations. However, scanning electron microscopy (SEM) conforms the fabrication-induced defect such as voids, interfacial delamination, misalignment and pinhole fault within via. Among them pinhole defects are severe as they compromise dielectric integrity, leakage path and may ultimately lead to TSV failure. Therefore, pinhole fault require special attention during defect modeling and reliability assessment of of via.

TABLE I: Geometrical and Material Parameters of the TSV

Symbol	Description	Value	Unit
P_{TSV}	Pitch between two TSVs	10	μm
d_{TSV}	TSV diameter	4.94	μm
r_{TSV}	TSV radius	2.47	μm
h_{TSV}	TSV height	15	μm
t_{ox}	Oxide layer thickness	0.13	μm
t_{dep}	Depletion layer thickness	0.87	μm
l_{sub}	Silicon substrate length	3.06	μm
σ_{Si}	Silicon conductivity	10	S/m
σ_{Cu}	Copper conductivity	5.9×10^7	S/m
ρ_{TSV}	Copper resistivity	1.68×10^{-10}	$\Omega \cdot \text{m}$
$\epsilon_{r,Si}$	Relative permittivity of silicon	11.9	—
$\epsilon_{r,ox}$	Relative permittivity of oxide	4	—
$\mu_{r,TSV}$	Relative permeability of TSV	1	—
δ_{ox}	Oxide loss tangent	0	—
δ_{Si}	Silicon loss tangent	0	—

Although the recent studies focused on the fabrication-induced defect, defect shape, size and location. The electrical modeling of these defects has received relatively limited attention. Therefore, Shang Y. *et al.*[], presented an electrical model for pinhole defected via by representing the defect as an equivalent leakage admittance and deriving analytical expression based on the pinhole geometry. However, the analysis is confined to the pinhole region itself whereas the effect of surrounding intact dielectric region on leakage behavior and log-term reliability remain unexplored. Long C. *et al.* [] and Yang Z. *et al.* [] investigated the effect of a cylindrical pinhole defect on the insertion loss and signal transmission characteristics of TSVs. However, the electrical behavior of the pinhole defect and its impact on the TSV performance have not been comprehensively analyzed.

The remainder of this paper is organised as follows. **Section II** describes the physical and electrical modelling of both pinhole-defected TSV cases. **Section III** demonstrates the mathematical equations used for the TSV models. **Section IV** presents the simulation setup and results, including the analysis of S-parameters and the performance analysis. Finally, **Section V** summarises and concludes the paper's objective that compares the performance of the different pinhole defect models and evaluates their impact on TSV reliability and signal integrity with a well defined summary.

II. TSV DESIGN AND DEFECT MODELLING

A pinhole is defined as a small, microscopic hole in the dielectric liner caused by fabrication imperfections, such as contamination trapped during dielectric layer deposition or non-uniform oxide layer deposition. This pinhole defect weakens the insulation between the TSV and the silicon substrate,

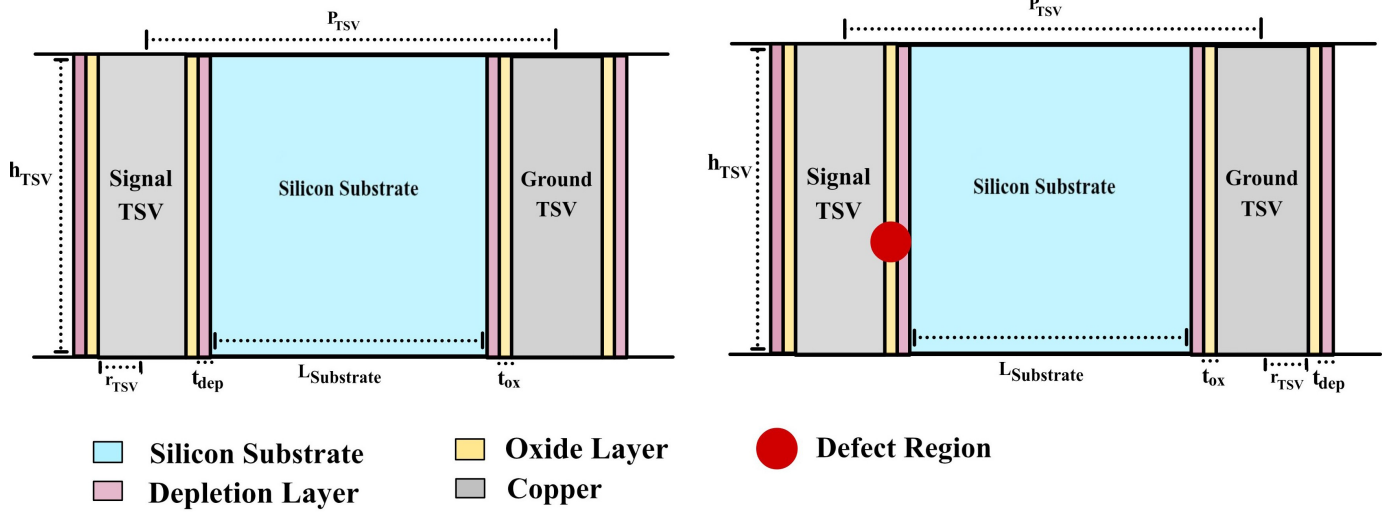


Fig. 1: Physical structure of a defect-free TSV and a pinhole-defected TSV

altering the electrical behaviour of the TSV. Pinhole defects are one of the most common dielectric liner defects in TSVs. This defect creates an unintended leakage path between the TSV and the silicon substrate. Consequently, there is an increase in leakage current and leakage power in the system. In addition, they degrade signal integrity by affecting transmission characteristics (noise, crosstalk, and propagation delay). The presence of these defects compromises the overall reliability of 3D ICs. The electrical behaviour of a TSV incorporating a pinhole defect depends on the material occupying the defect area. When the pinhole is filled with leaked copper, a conductive leakage path is created between the TSV and silicon substrate, which is modelled using an equivalent resistance and induction due to the leakage current. In contrast, when the pinhole region is an air-filled void, the effective dielectric constant changes, which results in a variation of the local TSV parasitic capacitance that is represented by an equivalent capacitance and conductance. To investigate the influence of pinhole defects on TSV performance, two defect scenarios are considered. For each scenario, the physical structure is first modelled, followed by the derivation of its corresponding electrical equivalent model.

A. Cu-Leaked Pinhole Defect Model

1) *Physical Modelling*: The physical model fig. 1, demonstrates the structural cross section of a pair of TSVs embedded within a silicon substrate. One TSV works as the signal path, while the other serves as a ground return path. Each TSV consists of a cylindrical copper conductor surrounded by a silicon dioxide (SiO_2) insulating liner, which electrically isolates the copper cylinder from the silicon substrate. These physical dimensions contribute in the electrical characteristics of the TSV, including its resistance, inductance, capacitance and conductance (RLGC). The red circle on the oxide layer represents the position of the pinhole defect for both the cases.

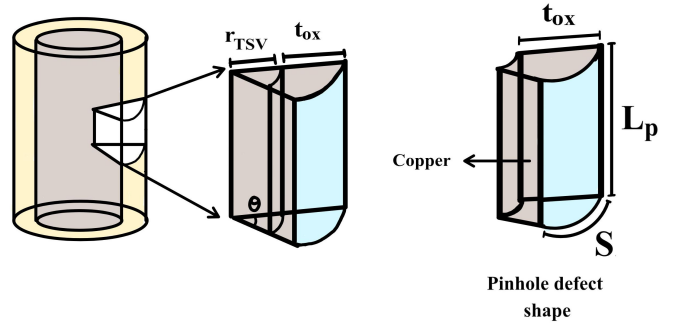


Fig. 2: Cu-leaked pinhole defect

Fig. 2 illustrates the close up physical 3 Dimensional of the copper leakage pinhole defect. The shape of the pinhole defect is taken as bent rectangular prism with angular extent (θ) radians and the length of the defect L_p (μm) with the bent length as S (pm). Furthermore, the formula for S is discussed in Section 03. Due to this defect a leakage resistance and leakage inductance are induced. Since copper is present inside the pinhole there will be resistance and due to this, an unintentional conductive path between the TSV and the silicon substrate, a small amount of leakage current will flow through it. Because of that leakage current, Leakage Inductance will be induced in the horizontal orientation.

2) *Electrical Model*: The equivalent electrical model of a cu-leaked pinhole-defected TSV is modelled and shown in Fig. 3. In this figure, the defect area is represented in terms of resistance and inductance formed due to that leakage path. The direction of resistance and leakage Induction taken corresponding to the leakage current flow from TSV to silicon substrate through the dielectric liner[6]. Here, the resistance of the defect is taken as copper resistance, since the defect region

TABLE II: Parameters Used for Pinhole Defect Modelling

Symbol	Description	Value	Unit
J	Leakage factor	40%	–
θ	Pinhole sector angle	$\pi/2$	rad
L_p	Pinhole sector length	4	μm
T	Operating temperature	298	K
T_0	Reference temperature	273	K
α	Temperature coefficient of resistivity	0.039	K^{-1}
ρ_0	Resistivity at reference temperature	1.68×10^{-8}	$\Omega\cdot\text{m}$
ρ_T	Resistivity at operating temperature	3.32×10^{-8}	$\Omega\cdot\text{m}$

is filled with copper.

B. Air Void Pinhole Defect

1) *Physical Modelling*: The structural cross sectional representation is the same as in the copper leakage case except here, instead of copper getting leaked into, the air will be present in the defect region. We considered that the length of the defect is the same as in the previous case L_p (m) and the angular extent (θ) radians and the bent width as s (pm) as well.

Fig. 4 illustrates the bent rectangular prism view of the pinhole defect. In this case the defect area represents the air void. Because of this type of defect case, a leakage conductance and leakage capacitance will be induced, altering the local TSV parasitic capacitance and conductance. The Leakage capacitance is because the air acts as a dielectric between the TSV (copper conductor) and silicon substrate.

2) *Electrical Model*: Fig. 5. In the presence of an air void defect within the dielectric liner the defect is modelled by an additional parallel RC branch comprising the leakage capacitance because of altered electric fields due to the air void and leakage conductance which is modelled by taking the inverse of the leakage resistance. The architectural positioning of the leakage conductance and capacitance within the equivalent circuit model is adopted from [7].

Fig. 5. In the presence of an air void defect within the dielectric liner the defect is modelled by an additional parallel RC branch comprising the leakage capacitance because of altered electric fields due to the air void and leakage conductance which is modelled by taking the inverse of the leakage resistance. The architectural positioning of the leakage conductance and capacitance within the equivalent circuit model is adopted from [7].

III. MATHEMATICAL FORMULATION

A. Defect Free TSV Parasitic Equations

This subsection presents the mathematical formulae of the defect-free TSV parasitic parameters based on its physical structure and material properties. The equivalent model is defined by the RLGC components, which together explain the electrical behaviour of the TSV. These parameters are analytically derived and serve as a baseline model for evaluating the impact of fabrication-induced defects in the following sections.

1) Resistance:

$$R_{\text{TSV}} = \sqrt{R_{\text{dc,TSV}}^2 + R_{\text{ac,TSV}}^2} \quad (1)$$

$$R_{\text{dc,TSV}} = \frac{\rho_{\text{TSV}} h_{\text{TSV}}}{\pi r_{\text{TSV}}^2} \quad (2)$$

$$R_{\text{ac,TSV}} = \frac{\rho_{\text{TSV}} h_{\text{TSV}}}{2\pi r_{\text{TSV}} \delta_{\text{SD}} - \pi \delta_{\text{SD}}^2} \quad (3)$$

$$\delta_{\text{SD}} = \frac{1}{\sqrt{\pi f \mu_{\text{TSV}} \sigma_{\text{TSV}}}} \quad (4)$$

Eq. (1) presents the parasitic resistance of the defect-free TSV model, which is modeled as a frequency-dependent parameter. Eq (2) The DC resistance R_{dc} dominates at low frequencies where the current is uniformly distributed across the conductor cross section whereas the Eq. (3) AC resistance R_{ac} becomes significant at high frequencies. This transition in resistance behaviour is primarily due to the skin effect. At high frequencies, the current tends to concentrate near the surface of the conductor. There by reducing the effective conducting area. This effect introduces the concept of the skin depth δ_{SD} which defines the depth up to which current effectively goes into the conductor and plays a pivotal role in determining the AC resistance.

2) Inductance:

$$L_{\text{TSV}} = \frac{1}{2} \left(\frac{\mu_0 \mu_{r,\text{TSV}} h_{\text{TSV}}}{2\pi} \right) \ln \left(\frac{P_{\text{TSV}}}{r_{\text{TSV}}} \right) \quad (5)$$

Eq. (5) models the defect-free TSV inductance. It quantifies the inductance produced by the magnetic flux linkage between the current flowing from the TSV and its return path in the surrounding conductive structure.

3) Conductance:

$$G_{\text{Si}} = \frac{\pi \sigma_{\text{Si}} h_{\text{TSV}}}{\cosh^{-1} \left(\frac{P_{\text{TSV}}}{d_{\text{TSV}}} \right)} \quad (6)$$

Eq. (6) describes the defect free conductance of the TSV. It models the leakage behavior TSV substrate material, a real life substrate does not act like ideal dielectric which have infinite insulating property, so silicon substrate will have some conductance in it. That conductance is this formula.

4) Capacitance:

$$C_{\text{ox}} = \frac{2\pi \epsilon_0 \epsilon_{r,\text{ox}} h_{\text{TSV}}}{\ln \left(\frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}} \right)} \quad (7)$$

$$C_{\text{dep}} = \frac{2\pi \epsilon_0 \epsilon_{r,\text{Si}} h_{\text{TSV}}}{\ln \left(\frac{r_{\text{TSV}} + t_{\text{ox}} + t_{\text{dep}}}{r_{\text{TSV}} + t_{\text{ox}}} \right)} \quad (8)$$

$$C_{\text{ins}} = \frac{C_{\text{dep}} C_{\text{ox}}}{C_{\text{dep}} + C_{\text{ox}}} \quad (9)$$

$$C_{\text{Si}} = \frac{\pi \epsilon_0 \epsilon_{r,\text{Si}} h_{\text{TSV}}}{\text{acosh} \left(\frac{P_{\text{TSV}}}{d_{\text{TSV}}} \right)} \quad (10)$$

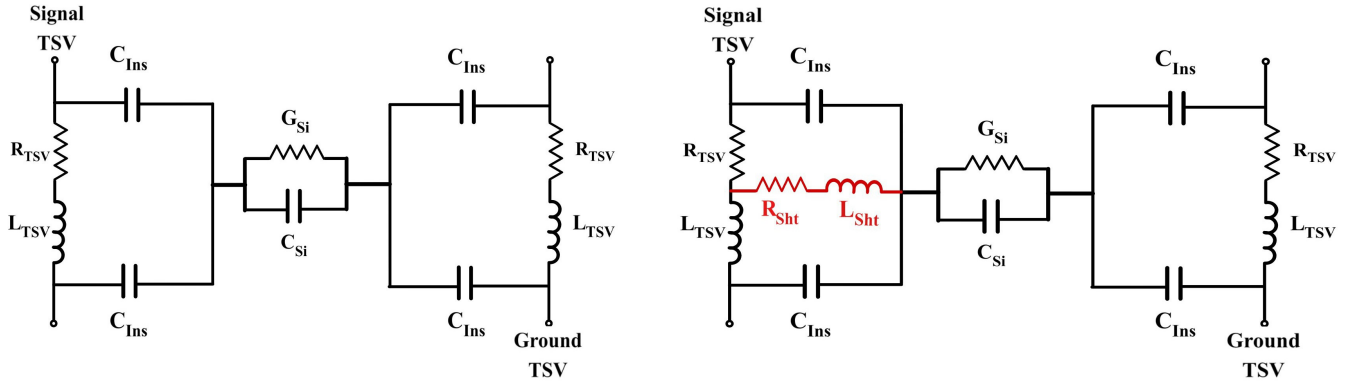


Fig. 3: Equivalent electrical model of a defect free TSV and Cu-leaked pinhole defected TSV

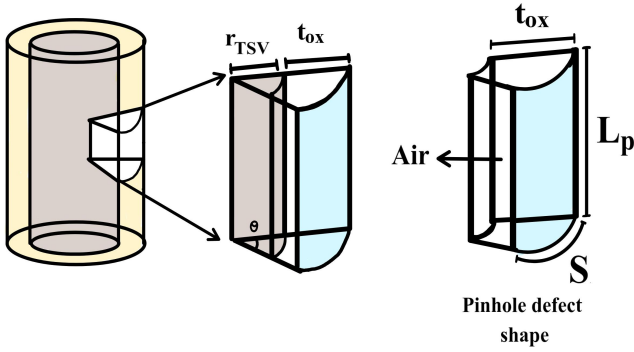


Fig. 4: Air void pinhole defect

The C_{ox} models the oxide layer capacitance, where SiO_2 acts like a dielectric medium separating two conductors (TSV and the silicon substrate) forming oxide capacitance. C_{dep} represents the depletion region capacitance formed in the silicon substrate due to the reduced charge carrier concentration near the TSV interface. C_{Si} accounts for the silicon substrate coupling effect between the adjacent TSVs or between the TSV and the return path. Finally, C_{ins} is the effective insulation capacitance obtained by taking the series combination the oxide and depletion capacitances. Together these capacitances define the overall electric field coupling behaviour of the TSV and play a crucial role in determining signal integrity and overall performance.

B. Copper-Filled Pinhole Mathematical Model (RL)

1) Short Defect Resistance:

$$R_{sht} = \frac{\rho_{Cu} t_{ox}}{L_p \theta (r_{TSV} + t_{ox})} \quad (11)$$

Short defect resistance represents the effective resistance created when an unintended conductive path is formed inside the TSV due to a manufacturing defect. This resistance modifies the normal current flow by introducing an additional low resistance leakage path parallel to the TSV conductive path. Thereby affecting the signal integrity and increases

leakage current. its value depends on the geometry of the defect material conductivity and the length of the unintended conductive region.

2) Short Defect Inductance:

$$L_{sht} = \frac{\mu_{TSV} t_{ox}}{2\pi} \left[\text{acosh} \left(\frac{S}{\sqrt{L_p^2 + S^2}} \right) - \sqrt{t_{ox}^2 + L_p^2 + S^2} + \frac{S}{4} + \sqrt{L_p^2 + S^2} \right] \quad (12)$$

Short defect inductance represents the additional inductive effect by the defect-induced current loop inside the TSV structure. When a short defect is present, the current redistributes and forms an altered return path which increases the magnetic field coupling in the structure. This results in an additional inductive component that depends on the defect structure, permeability of the material involved in that defect and also the effective loop area created by the short path. This parasitic inductance impacts high frequency behaviour leading to the changes in delay and crosstalk performance.

C. Air-Void Pinhole Mathematical Model (CG)

1) Short Defect Capacitance:

$$C_{air} = \frac{\theta \epsilon_0 \epsilon_{r,air} L_p}{\ln \left(\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right)} \quad (13)$$

$$C_{SiO_2,remain} = \frac{(2\pi - \theta) \epsilon_0 \epsilon_{r,ox} L_p}{\ln \left(\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right)} \quad (14)$$

$$C_{short} = C_{air} + C_{SiO_2,remain} \quad (15)$$

Short defect capacitance represents the additional capacitive coupling introduced due to the presence of the air void defect pinhole within the TSV structure which acts like the dielectric between the TSV and the substrate creating capacitance between them. The defect modifies the local electric field distribution effectively increasing the coupling between the TSV conductor and the surrounding silicon or adjacent structures.

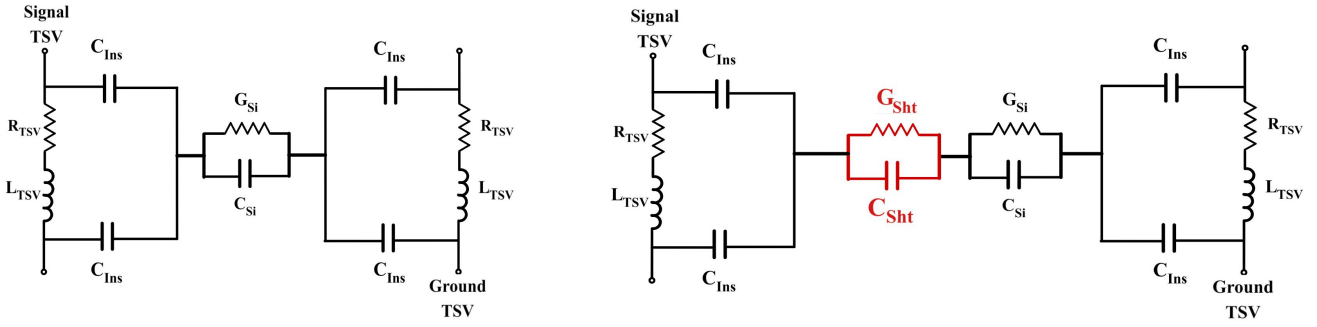


Fig. 5: Equivalent electrical model of a defect-free and air-void pinhole-defected TSV

This additional capacitance also affects the local capacitances as well as high frequency performance by increasing signal delay and crosstalk.

This formula's capacitance is derived by taking the whole cylinder part with height of L_p . That cylinder part is divided into the sector with air and the remaining sector with SiO_2 according to it C_{air} and $C_{\text{SiO}_2, \text{remain}}$ are derived from standard operations. Now the short defect is defined as the addition of these two capacitances ($C_{\text{SiO}_2, \text{remain}} + C_{\text{air}}$).

2) Short Defect Conductance:

$$G_{\text{air}} = \frac{\theta \varepsilon_0 \varepsilon_{r, \text{air}} L_p f}{\ln\left(\frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}}\right)} \quad (16)$$

$$G_{\text{SiO}_2, \text{remain}} = \frac{(2\pi - \theta) \sigma_{\text{SiO}_2} L_p}{\ln\left(\frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}}\right)} \quad (17)$$

$$G_{\text{short}} = G_{\text{SiO}_2, \text{remain}} + G_{\text{air}} \quad (18)$$

Short defect conductance represents the additional leakage conductance introduced when a defect creates the leakage path. This occurs due to the formation of low resistance[7]. This conductance is strongly dependent on the defect material properties, geometry and contact area. similar to the short capacitance, the short conductance is also taken as the addition of two conductances ($G_{\text{SiO}_2, \text{remain}} + G_{\text{air}}$).

IV. SIMULATION METHODOLOGY AND PERFORMANCE EVALUATION

The following subsections describe the implementation performance analysis and S-parameters results of the proposed models obtained using the softwares. For the performance analysis Hspice is used for all three models to calculate noise, propagational delay, crosstalk and leakage power, leakage current. The equivalent RLGC parameters derived from the analytical formulations are used to construct the electrical model in Cadence for circuit level analysis of S-parameters. For electromagnetic validation a 3D physical TSV structure is modelled in HFSS to capture the field distribution and frequency-dependent behaviour. Both defect-free and defective cases are simulated under identical boundary conditions to ensure fair comparison.

A. Leakage Current and Leakage Power

In terms of leakage performance, Fig. 9 both leakage current and leakage power in the proposed work scale upward when moving away from the defect-free optimised model. In this performance parameter the proposed RL models show 3.29% reduced leakage for both current and power than [6] and 5.47% reduced leakage in the CG model than [7].

B. Noise

The performance parameter Fig. 6 compares the noise characteristics of the proposed models with the corresponding reference model reported in the literature. The noise voltage of the proposed design peaks at the intermediate RL model at approximately $3.46 \mu\text{V}$ before dropping down to $3.21 \mu\text{V}$ in the CG Model. The defect-free model achieves the lowest overall noise floor. Throughout all configurations, the proposed RL model shows superior noise resistance keeping the noise profile 9.03% lower than [6] and 6.06% lower than [7] in the CG model. The defect-free TSV exhibits the lowest noise level, indicating that the dielectric liner completely isolates the TSV reducing current leakage and unwanted electrical disturbances. In the absence of defect-induced parasitic elements the signal transmission is governed only by TSV resistance, TSV inductance and substrate parasitics. resulting in a stable current distribution and reduced noise generation. In contrast, the pinhole-defected TSVs exhibit increased noise levels due to the introduction of additional defect-induced parasitic elements that disturb the normal signal propagation.

The Copper leaked pinhole defect produces the highest noise among the others because of the additional short defect resistance and the short defect inductance branch introduces an unintended conductive path inside the structure. This metallic leakage path causes the signal current to redistribute between the intended path and the unintended path. The increased parasitic impedance produces additional thermal and inductive noise. Consequently, Larger electrical fluctuations are introduced to the signal transmission. On the other hand the air void model establishes shunt elements short defect capacitance and short defect conductance which primarily increases the capacitive coupling and limited dielectrical leakage rather than a direct conductive path. AS a result, the CG model exhibits a

TABLE III: Comparison of Defect-Free and Defective TSV Parasitic Parameters for 15 μm via height

Parameter	Defect-Free TSV	Air-Void Pinhole TSV	Cu-Leaked Pinhole TSV
R_{TSV} (m Ω)	82.8	82.8	82.8
L_{TSV} (pH)	2.09	2.09	2.09
C_{Ins} (fF)	11.55	11.25	11.55
C_{Ox} (fF)	65.07	61.80	65.07
C_{Dep} (fF)	35.84	34.40	35.84
C_{Sub} (fF)	3.73	2.75	3.73
G_{Sub} (μS)	354	273	354
R_{Sht} (m Ω)	–	–	0.133
L_{Sht} (H)	–	–	1.90×10^{-20}
C_{Sht} (fF)	–	14.09	–
G_{Sht} (μS)	–	108	–

lower noise level than the RL model. The only leakage we see in the air void pinhole model is displacement current which is very small compared to the leakage current in the cu-leaked model.

C. Delay

Regarding the total propagational delay fig. 7. The proposed structure achieves a steady linear increase across the three models. The curve starts with its lowest point under 0.49 fs for the defect-free model and then rises to the RL model and finally reaches the CG model at its highest point of 0.54 fs. The proposed RL model demonstrates a 5.09% reduced propagational delay than [6] and 9.25% than [7] in the CG model.

For a phase dependent propagation delay, Fig. 8 demonstrates the proposed design an increase in both inphase and outphase delays when moving from the defect-free to the RL and CG models. At the RL stage the proposed work shows 7.89% reduced crosstalk inphase and 3.07% reduced crosstalk outphase than [6]. And for CG stage the proposed work shows 9.09% reduced crosstalk inphase and 7.20% reduced crosstalk out phase than [7]. The graph compares the propagational delay of the models. Defect-free exhibits the lowest propagational delay due to the absence of the defect-induced parasitic elements. The cu-leaked pinhole model has a moderate delay because of the redistribution of signal current. Thereby slowing the signal propagation. But in air void pinhole model the delay is highest because of the additional short defect capacitance and conductance. As the capacitive load increases it requires additional charging and discharging time. Fig. 8. Crosstalk analysis performed under both inphase (pulse excited signal TSV and a grounded TSV) and outphase (Two pulse excited TSVs) conditions. The results indicate that defective TSVs experience a higher coupling effect compared to the defect-free case. The air void defect exhibits stronger capacitive coupling because of the short defect capacitance (additional charging and discharging time) leading to increased crosstalk. In contrast, the copper leaked pinhole model primarily affects resistive and inductive coupling resulting in moderate crosstalk delay.

D. S Parameters

The S parameters results obtained using two different software packages, one is Cadence and the other one is HFSS, are then plotted in Figs. 10,11. The S_{11} parameter increases in defective cases, indicating higher reflection due to the impedance mismatch. Similarly, S_{21} decreases showing reduced transmission efficiency as we go from defect-free to defective TSV models. The defect-free TSV has only the intended parasitic RLGC so its input impedance is closer to the desired characteristic impedance. As the impedance mismatch is small, most of the signal is transmitted through the TSV and compared to defect models the S_{21} parameter most of the signal transmitted to the output node. On the other hand, in the defective models, the additional parasitic parameters will be added, which will lead to the impedance match. This will cause the maximum signal to get reflected and most of the signal is lost during transmission.

E. Eye Diagrams

Fig. 12. presents the eye diagrams of the defect-free TSV, and other two defected TSV models. The defect-free TSV exhibits the largest eye opening with the highest eye height and the lowest jitter indicating excellent signal integrity. The introduction of pinhole defect reduces the eye height and increases the jitter due to the additional parasitic effects introduced by the defects. Among the defective models the air void pinhole defect exhibits the smallest eye opening and highest jitter, demonstrating the most significant degradation in signal integrity.

V. CONCLUSION

The paper presented the modelling and performance analysis of defect-free and pinhole-defective TSVs, considering two practical fabrication-induced defect mechanisms Cu-leaked pinhole and air-void pinhole. The corresponding electrical equivalent models were developed by including the defect-induced parasitic elements and their impact on TSV performance was evaluated through HSpice, Cadence and HFSS simulations.

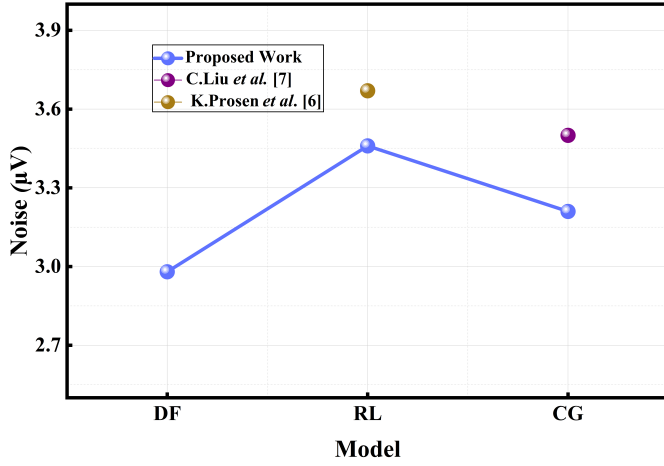


Fig. 6: Noise performance comparison between the proposed and reference TSV models

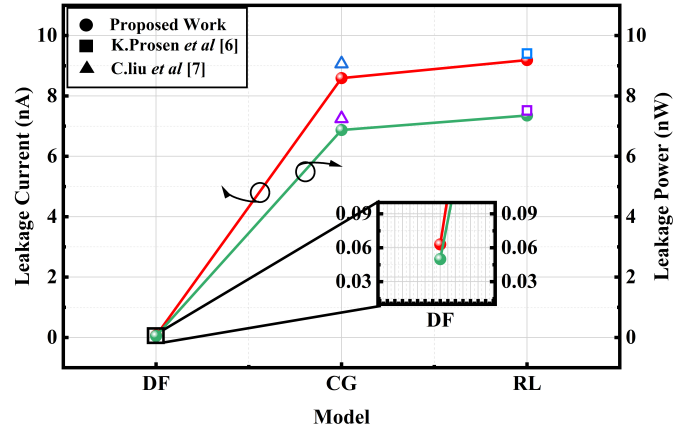


Fig. 9: Comparison of leakage characteristics between the proposed and reference TSV models.

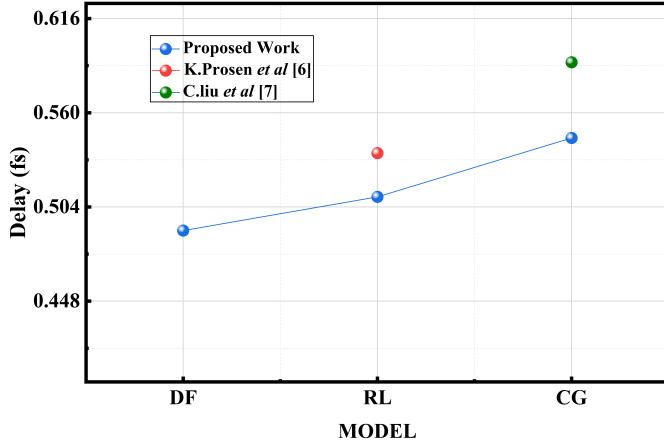


Fig. 7: Delay performance comparison between the proposed and reference TSV models.

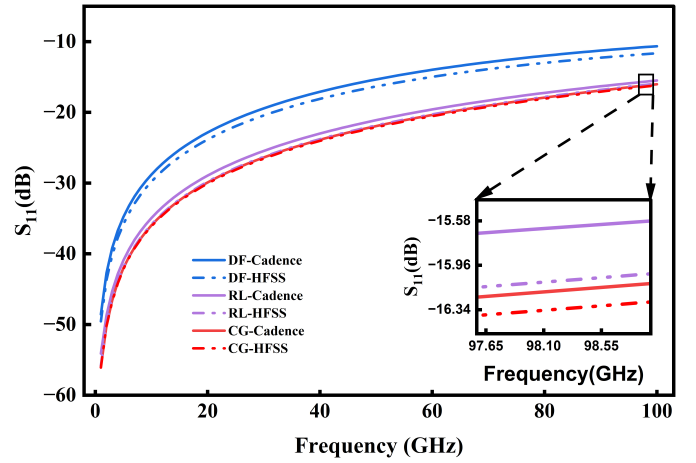


Fig. 10: S_{11} parameter comparison of defect-free and defective TSV models

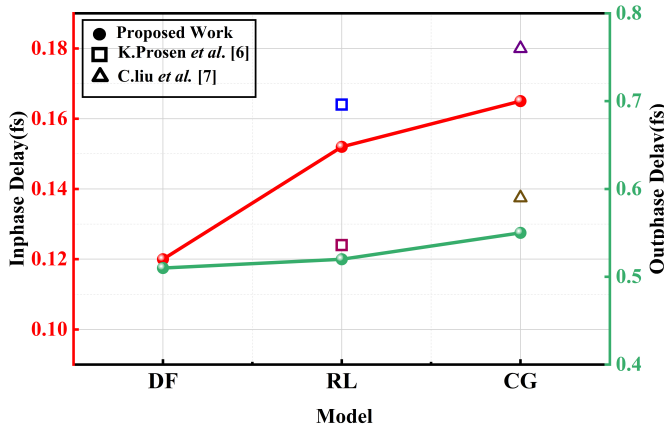


Fig. 8: Comparison of crosstalk characteristics between the proposed and reference TSV models.

The simulation results demonstrate that fabrication defects significantly degrade TSV electrical performance. Among the proposed defect models, the cu-leaked defect exhibits the highest noise level and leakage current. In contrast, the air void pinhole defect produces the highest propagation delay and crosstalk. Moreover, both defective models exhibit increased reflection and reduced transmission.

From a reliability perspective, the cu-leaked pinhole defect is more critical because it establishes a direct conductive leakage path through the dielectric liner. Which can lead to increased leakage current, excessive power dissipation, local heating and eventual TSV failure. Therefore, preventing copper leakage during fabrication is essential for improving longterm TSV reliability. From a signal integrity perspective, the air void pinhole defect is more concerning because its increased capacitive coupling, propagation delay and crosstalk degrade high speed signal transmission. Making it a dominant concern in maintaining signal quality in 3D ICs.

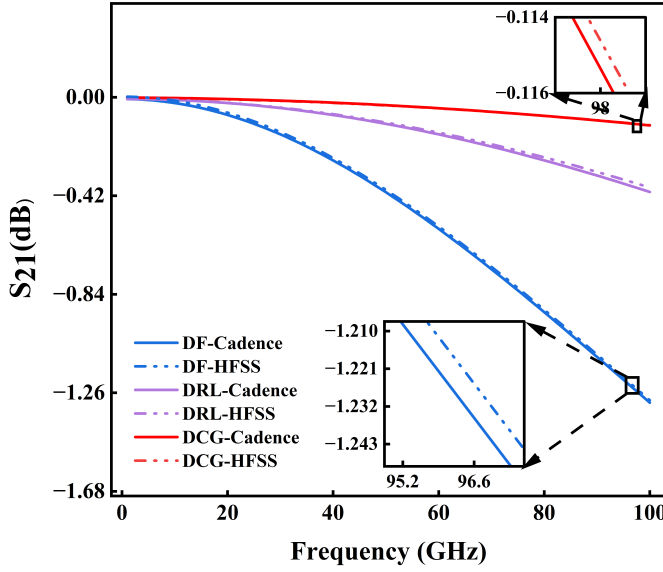


Fig. 11: S_{21} parameter comparison of defect-free and defective TSV models

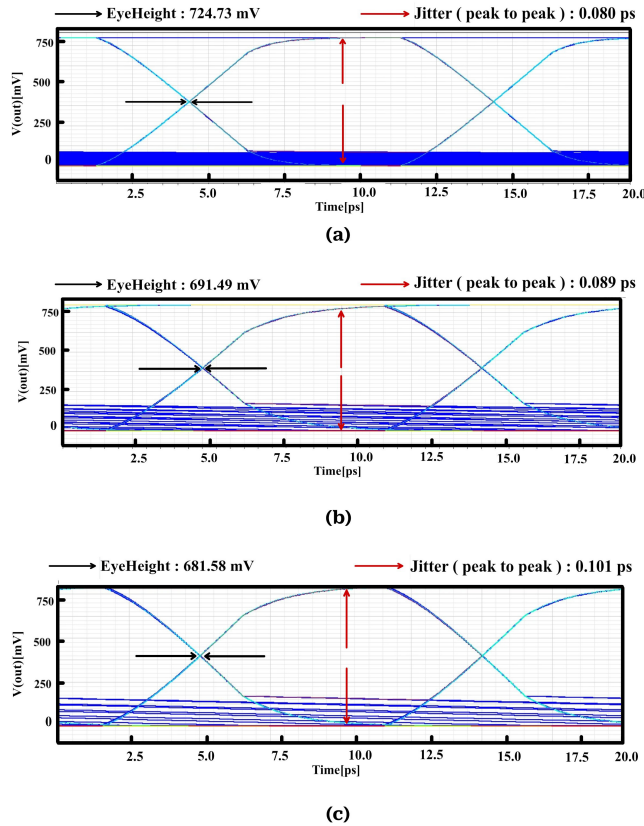


Fig. 12: (a) Defect-free TSV model eye diagram, (b) Cu-leaked pinhole defective TSV model eye diagram, (c) air void pinhole defective TSV model eye diagram

Another important contribution of this work is the consideration of defect geometry. Unlike the previous studies, where different defects were presented using different geometries this work models both cu-leaked and air void pinhole defects using the same bent rectangular prism geometry. Maintaining identical defect geometry eliminates shape dependent variations allowing the observed electrical characteristics to be governed by defect mechanism rather than differences in physical representation. This demonstrates that defect geometry itself influences the resulting parasitic parameters and subsequently affects both signal integrity and reliability analysis. Therefore, accurate geometry representation of fabrication defects is essential for developing realistic TSV models and for predicting the electrical behaviour of practical 3D ICs.

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